

CSC 380 — Quiz 04 Practice Guide

Banner — PUBLISHED SCOPE; due date NEEDS VERIFICATION.

- **Scope is published** (D2L announcement, Sep 18, 2026): Bayes' rule, the law of total probability (LOTP), and independence, covering **slides pp. 43–68**.
- **The due date is not verified.** Quiz 04 was not on Gradescope as of the 2026-09-24 audit (~14:03 Arizona time). The syllabus cadence ("quizzes every Tuesday unless noted") is **not** a verified due date. **Never invent a due date.** This is an open practice guide, not a countdown plan.

Canonical maps: csc380-topic-map.md * announced-assessments-inventory-2026-09-24.md

1. Header

Field	Value	Status
Course	CSC 380 Principles of Data Science (FA26), TuTh 5:00–6:15 PM, M Pacheco ILC 130	Confirmed
Assessment	Quiz 04 (practice guide)	Item announced
Due date/time	Needs verification — do not invent	NV
Coverage	Bayes' rule; LOTP; independence; slides 43–68	PUBLISHED SCOPE
Quiz policy (syllabus)	12 quizzes, each 1.5% of the grade; the two lowest are dropped; no make-ups	Confirmed (syllabus)
Past quiz format	Quiz 01 was one graded rubric question on a one-page submission, worth 1.5 points	Practice recommendation (Gradescope notes)
Guide updated	2026-09-24 (America/Phoenix) — Opus + Content-fold	

All times are **Arizona time (MST)**.

Sources (Content-pass 2026-09-24)

Source	Role
D2L announcement (Sep 18): Quiz 04	PUBLISHED SCOPE: Bayes, LOTP, independence; slides pp. 43-68
Content Resources/quiz01_03.pdf	Practice-pack grain (Quiz 01-03 style). Quiz 03 is LOTP+Bayes; used to shape Practice recommendation items below. Not a live quiz attempt.
Content Homework/homework02.pdf, homework03.pdf	Overlapping probability practice context (HW3 due Tue Sep 29, 11:59 PM). Do not treat this guide as HW solutions.
Content Homework/homework01.zip	Earlier pandas/data practice; not Quiz 04 core.
Gradescope format notes	One-page / 1.5-pt quiz habit (Quiz 01)

Related but separate: HW3 (Probability 2) is due **Tue Sep 29, 11:59 PM**. It has **two** Gradescope items, HW03 (written) and HW03 Code, and both must be submitted. It overlaps these topics, but HW3 is **not** this guide's target, and nothing here solves it.

Notation. In this guide, **not-A** means the complement of A (often written not-A or not-A on slides). $P(A | B)$ is read "probability of A given B."

2. Topic checklist (learning order)

Order	Section ID	Topic	Blind Y	Check Y	Retake Y
1	380-Q04-refresh	Conditional probability refresher			
2	380-Q04-lotp	Law of total probability			
3	380-Q04-bayes	Bayes' rule			
4	380-Q04-indep	Independence (vs. mutually exclusive)			
5	380-Q04-slides	Work the examples on slides 43–68			

Alias: 380-L-prob3 $\leadsto$ 380-Q04-bayes, for calendar continuity.

3. Plain explanations + worked examples

Conditional probability refresher (380-Q04-refresh)

$P(A | B) = P(A \text{ and } B) / P(B)$, when $P(B) > 0$.

Rearranged, this gives the **multiplication rule**: $P(A \text{ and } B) = P(A | B)P(B)$.

The Gradescope scores show the week-2 quiz (conditional probability) was the weakest of the three, so this refresher comes first.

Law of total probability (380-Q04-lotp)

If $B_1, \dots, B_k$ **partition** the sample space (they don't overlap and together cover everything):

$$P(A) = \sum_i P(A | B_i)P(B_i)$$

With two parts: $P(A) = P(A | B)P(B) + P(A | \text{not-}B)P(\text{not-}B)$.

The intuition: split the world into cases, find A's probability in each case, and weight each case by how likely it is.

Bayes' rule (380-Q04-bayes)

$P(B_j | A) = P(A | B_j) \cdot P(B_j) / P(A)$, where $P(A)$ comes from LOTP.

Table method (it makes normalizing automatic):

Hypothesis	Prior	Likelihood $P(\text{data} \mid H)$	Prior x likelihood	Posterior
...	...	...	...	(row value) / (column sum)

Worked example (disease test). $P(D) = 0.01$, $P(+ \mid D) = 0.9$, and $P(+ \mid \text{not-}D) = 0.05$.

- $P(+) = 0.90 \cdot 0.01 + 0.050 \cdot 0.99 = 0.009 + 0.0495 = 0.0585$.
- $P(D \mid +) = 0.009 / 0.0585 \approx \mathbf{0.154}$.

The disease is rare (a low **base rate**), so most positive results are false positives, even though the test is fairly accurate.

Independence (380-Q04-indep)

A and B are independent when $P(A \text{ and } B) = P(A) \cdot P(B)$. Equivalently, $P(A \mid B) = P(A)$ (when $P(B) > 0$), or $P(B \mid A) = P(B)$.

- **Coin example.** For two fair flips, $P(H1 \text{ and } H2) = 1/4 = (1/2)(1/2)$, so the flips are independent.
- **Mutually exclusive is not independent.** If $P(A) > 0$ and $P(B) > 0$ but they cannot both happen, then $P(A \text{ and } B) = 0 \neq P(A) \cdot P(B)$. Knowing A happened tells you B did not, which is maximal dependence.
- To **check** independence, compute both sides of the product rule. Don't rely on intuition.

4. Practice questions (core)

Recall

- R1. Write Bayes' rule for hypotheses H and not-H, given data D.
- R2. State independence in three equivalent ways.

Solve

- S1. Three urns are equally likely: U1 has 2 red and 1 blue, U2 has 1 red and 1 blue, U3 has 0 red and 2 blue. You pick an urn at random and draw one ball, which is red. What is $P(U1 \mid \text{red})$?
- S2. $P(A) = 0.3$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.12$. Are A and B independent?
- S3. $P(A \mid B) = 0.5$, $P(B) = 0.2$, $P(A \mid \text{not-}B) = 0.1$. Find $P(A)$ and $P(B \mid A)$.

Trace

- T1. For S1, fill in the Bayes table: prior, likelihood, prior x likelihood, posterior.

Explain

- E1. Why does LOTP appear in the denominator of Bayes' rule?
- E2. What is the difference between independent and mutually exclusive?

Code (optional study aid)

- C1. Simulate about 10,000 draws of the S1 setup in Python. Compare the fraction of red draws that came from U1 with the exact answer.

4b. One-page mock quizzes (1.5 points each) — Practice recommendation

Where this format comes from. Quiz 01's Gradescope result was one rubric question on a one-page submission, worth 1.5 points. Each mock below is **one multi-part problem that fits on one page**, with three parts worth 0.5 points each. The Quiz 04 interface and due date are **not known**. Give each mock **10 minutes**, closed-notes.

Mock Quiz A — LOTP + Bayes (factory)

Machine M1 makes 60% of the items, and 2% of its items are defective. Machine M2 makes the other 40%, and 5% of its items are defective.

- (a) [0.5] Find $P(\text{defective})$.
- (b) [0.5] An item is defective. Find $P(M2 \mid \text{defective})$.
- (c) [0.5] In one sentence, explain why (b) is larger than 0.4.

Mock Quiz B — Independence (dice)

Roll two fair six-sided dice. Let A = "the first die shows 6", B = "the sum is 7", and C = "the sum is 12".

- (a) [0.5] Are A and B independent? Show the product check.
- (b) [0.5] Are A and C independent? Show the product check.
- (c) [0.5] $P(E) = 0.2$, $P(F) = 0.5$, and E and F are mutually exclusive. Are they independent?

Mock Quiz C — Table -> conditional -> independence

Of 200 students, 80 drink coffee. 50 of the coffee drinkers are night owls, and 40 of the non-coffee drinkers are night owls.

- (a) [0.5] Find $P(\text{night owl})$.
- (b) [0.5] Find $P(\text{coffee} \mid \text{night owl})$.
- (c) [0.5] Are "coffee" and "night owl" independent? Justify with numbers.

Mock Quiz D — Bayes (spam filter)

$P(\text{spam}) = 0.3$. The word "free" appears in 40% of spam emails and in 5% of non-spam emails.

- (a) [0.5] Find $P(\text{"free"})$.
- (b) [0.5] Find $P(\text{spam} \mid \text{"free"})$.
- (c) [0.5] Explain in one sentence the mistake in saying " $P(\text{spam} \mid \text{free}) = 0.4$ ".

Quick checks

- **Qmu1.** $P(A) = 0.5$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.1$. Independent? Give a one-line reason.
- **Qmu2.** Write LOTP for an event D with the partition $\{B, \text{not-}B\}$.
- **Qmu3.** $P(H) = 0.2$, $P(E \mid H) = 0.9$, $P(E \mid \text{not-}H) = 0.3$. Find $P(H \mid E)$, showing the LOTP denominator.
- **Qmu4.** True or false: mutually exclusive events with positive probabilities are independent.
- **Qmu5 (stretch).** Using the disease test from section 3, a second, independent test is also positive. Assume the two results are independent given disease status. Find $P(D \mid \text{both positive})$.

4c. Content-sourced similar-style practice — Practice recommendation

Where this comes from. Content Resources/quiz01_03.pdf is a practice pack (not a started quiz). Its Quiz 03 item is LOTP then Bayes on two groups with different rates — the same published Quiz 04 skills. The items below are **new**

similar-style drills (numbers and story changed). Quiz 04 due date remains **NV**. Published scope is still Bayes / LOTP / independence only.

CQ1 — LOTP + Bayes (two clubs; Quiz 03 grain)

Two student clubs. Chess has 120 members; Debate has 80. Among Chess members, 25% always bring snacks to meetings. Among Debate members, 55% always bring snacks.

- (a) What fraction of all club members bring snacks?
- (b) A randomly selected student is bringing snacks. What is $P(\text{Debate} \mid \text{snacks})$?
- (c) In one sentence, why is (b) larger or smaller than Debate's membership share ($80/200 = 0.4$)?

CQ2 — Independence product check (HW3 grain)

Fair six-sided die. Let $A = \{2, 4, 6\}$ (even) and $B = \{1, 2, 3, 4\}$.

- (a) Compute $P(A)$, $P(B)$, and $P(A \text{ and } B)$. Are A and B independent?
- (b) Now let $A_2 = \{2, 3, 4, 6\}$ and keep B. Are A_2 and B independent?

CQ3 — Bayes (base-rate; HW3 shy-scientist grain)

At a large gathering, 70% of people are CS majors and 30% are data-science (DS) majors. 40% of CS majors own a mechanical keyboard; 80% of DS majors own one. You meet someone who owns a mechanical keyboard.

- (a) Find $P(\text{keyboard})$.
- (b) Find $P(\text{CS} \mid \text{keyboard})$.
- (c) Name the mistake in saying " $P(\text{CS} \mid \text{keyboard}) = 0.4$ ".

CQ4 — One-page mock (Quiz 01 format habit)

Factories F1 and F2 make 70% and 30% of chargers. Defect rates: 1% from F1, 4% from F2. A charger is defective.

- (a) [0.5] $P(\text{defective})$
- (b) [0.5] $P(F2 \mid \text{defective})$
- (c) [0.5] Are "defective" and "from F2" independent? Justify with one product check (you may use (a)-(b)).

5. ANSWER KEY (separate — check after a blind attempt)

Core keys

R1. $P(H | D) = P(D | H)P(H) / [P(D | H)P(H) + P(D | \text{not-H})P(\text{not-H})]$

R2. $P(A \text{ and } B) = P(A)P(B)$; $P(A | B) = P(A)$; $P(B | A) = P(B)$. The last two require the conditioning event to have positive probability.

S1. $P(\text{red}) = (2/3)(1/3) + (1/2)(1/3) + 0 = 2/9 + 1/6 = 7/18$. So $P(U1 | \text{red}) = (2/9) / (7/18) = 4/7$.

S2. $0.3 \times 0.4 = 0.12$, which equals $P(A \text{ and } B)$. **Yes, independent.**

S3. $P(A) = 0.50.2 + 0.10.8 = 0.18$. $P(B | A) = 0.10 / 0.18 = 5/9$.

T1.

Urn	Prior	Likelihood	Prior x likelihood	Posterior
U1	1/3	2/3	$2/9 = 4/18$	4/7
U2	1/3	1/2	$1/6 = 3/18$	3/7
U3	1/3	0	0	0
Sum	1		7/18	1

E1. The denominator is $P(\text{data})$, the total probability of the data across all hypotheses. LOTP computes exactly that, and dividing by it makes the posteriors sum to 1.

E2. Mutually exclusive means $P(A \text{ and } B) = 0$. Independent means $P(A \text{ and } B) = P(A)P(B)$, which is usually greater than 0. With positive probabilities, the two properties cannot both hold.

C1. The empirical fraction should land near $4/7 \approx 0.571$, within about ± 0.02 for 10,000 draws.

Content-sourced keys (CQ1-CQ4)

CQ1.

- (a) $P(\text{snacks}) = 0.25(120/200) + 0.55(80/200) = 0.250.6 + 0.550.4 = 0.15 + 0.22 = 0.37$.
- (b) $P(\text{Debate} | \text{snacks}) = (0.55 \times 0.4) / 0.37 = 0.22 / 0.37 = 22/37 \approx 0.595$.
- (c) Debate members bring snacks at a higher rate, so conditioning on snacks shifts belief toward Debate (from 0.4 up to ~ 0.595).

CQ2.

- (a) $P(A)=3/6=1/2$. $P(B)=4/6=2/3$. $P(A \text{ and } B)=P(\{2,4\})=2/6=1/3$. Product $(1/2) \times (2/3)=1/3$. **Independent.**
- (b) $P(A2)=4/6=2/3$. $P(A2 \text{ and } B)=P(\{2,3,4\})=3/6=1/2$. Product $(2/3) \times (2/3)=4/9 \neq 1/2$. **Not independent.**

CQ3.

- (a) $0.40.7 + 0.80.3 = 0.28 + 0.24 = 0.52$.
- (b) $0.28 / 0.52 = 7/13 \approx 0.538$.
- (c) 0.4 is $P(\text{keyboard} | \text{CS})$, the likelihood, not the posterior $P(\text{CS} | \text{keyboard})$.

CQ4.

- (a) $0.70.01 + 0.30.04 = 0.007 + 0.012 = 0.019$.
- (b) $0.012 / 0.019 = 12/19 \approx 0.632$.

- (c) $P(\text{defective and } F2) = 0.012$. $P(\text{defective})P(F2) = 0.0190.3 = 0.0057 \neq 0.012$. **Not independent.**

Mock quiz keys

Quiz A.

- (a) $0.60.02 + 0.40.05 = 0.012 + 0.020 = \mathbf{0.032}$.
- (b) $0.020 / 0.032 = \mathbf{0.625}$.
- (c) M2 has the higher defect rate, so seeing a defect shifts belief toward M2, from the prior of 0.4 up to 0.625.

Quiz B.

- (a) $P(A) = 1/6$. $P(B) = 6/36 = 1/6$. $P(A \text{ and } B) = P(\text{first die 6, second die 1}) = 1/36$, and $(1/6)(1/6) = 1/36$. **Independent.**
- (b) $P(C) = 1/36$. $P(A \text{ and } C) = 1/36$, but $P(A)P(C) = 1/216$. **Not independent.**
- (c) $P(E \text{ and } F) = 0$, but $P(E)P(F) = 0.1$. **Not independent.**

Quiz C.

- (a) $(50 + 40) / 200 = \mathbf{0.45}$.
- (b) $50 / 90 = \mathbf{5/9} \approx \mathbf{0.556}$.
- (c) $P(\text{owl} | \text{coffee}) = 50/80 = 0.625$, which is not $P(\text{owl}) = 0.45$. Equivalently, $P(\text{coffee and owl}) = 0.25$, but $P(\text{coffee})P(\text{owl}) = 0.40.45 = 0.18$. **Not independent.**

Quiz D.

- (a) $0.40.3 + 0.050.7 = 0.12 + 0.035 = \mathbf{0.155}$.
- (b) $0.12 / 0.155 \approx \mathbf{0.774}$.
- (c) 0.4 is $P(\text{free} | \text{spam})$, the likelihood. Treating it as $P(\text{spam} | \text{free})$, the posterior, ignores the prior and the base rate.

Quick-check keys

- **Qmu1.** No. $0.5 \cdot 0.4 = 0.2 \neq 0.1$.
- **Qmu2.** $P(D) = P(D | B)P(B) + P(D | \text{not-}B)P(\text{not-}B)$.
- **Qmu3.** $P(E) = 0.90.2 + 0.30.8 = 0.42$. $P(H | E) = 0.18 / 0.42 = \mathbf{3/7} \approx \mathbf{0.429}$.
- **Qmu4.** False. Their intersection has probability 0, but the product of their probabilities is greater than 0.
- **Qmu5.** $0.810.01 / (0.810.01 + 0.0025 \cdot 0.99) = 0.0081 / 0.010575 \approx \mathbf{0.766}$. Equivalently, use 0.154 as the new prior and update once more.

6. Common mistakes + readiness checklist

- Confusing $P(A | B)$ with $P(B | A)$ (base-rate neglect).
- Forgetting to normalize, i.e. skipping the LOTP denominator.
- Treating mutually exclusive events as independent.
- Using a "partition" whose parts overlap or leave gaps.
- **Inventing a Quiz 04 due date** from the Tuesday cadence.

Ready if:

- ☐ Mock Quizzes A-D are each done in 10 minutes or less with at least 1.0 of 1.5 points
- ☐ CQ1-CQ4 (Content-sourced) done blind, then checked
- ☐ Three independence checks are done by computing both sides
- ☐ The examples on slides 43–68 are worked
- ☐ **No countdown cram** until the due date is verified

7. Open practice checklist (no due-date cram plan)

Mode	Action
Anytime	Rotate short sets for 380-Q04-refresh, lotp, bayes, and indep
Alongside HW3 (due Sep 29, 11:59 PM)	Work these topics as separate practice. Upload both HW03 and HW03 Code.
When the due date is posted on Gradescope or D2L	Add a light review 24–48 hours before. Do not invent the date now.
Tuesdays and Thursdays	Keep light around classes. Do deep drills Mon/Wed/Fri and weekends.

8. Changes in this review (2026-09-24)

- Kept the **PUBLISHED SCOPE** and **due-date NV** banners unchanged in meaning.
- Replaced the complement symbol "c" with "not-A", because it rendered as a missing-glyph box in the old PDF. Added a notation note.
- Added quiz policy facts from the syllabus (12 quizzes, drop the 2 lowest, no make-ups) and the HW3 two-item submission reminder.
- Added Mock Quizzes A–D (one page, three parts at 0.5 points each), a filled Bayes table (T1), and a stretch sequential-update item. All have keys.
- Stated the weak-early-quiz signal in words only, with no scores, because the repository is public.
- Changed times to Arizona time.

8b. Content-fold (2026-09-24, America/Phoenix)

- Folded Content quiz01_03 + HW02/HW03 citations into Sources; added CQ1-CQ4 similar-style practice (**Practice recommendation**) from Quiz 03 / HW3 grain without copying live quiz or HW solutions.
- Published Quiz 04 scope unchanged (Bayes / LOTP / independence; slides 43-68). Due date still **NV**.
- ASCII-only math pass (no unicode subscripts / Sigma / mu) so PDF does not render missing-glyph boxes.

9. Study-aid / AI-policy disclaimer

Study aid only — not for submitted work (Quiz 04, HW, exams). Follow CSC 380 and UA academic-integrity and AI policies.